

Dividend Coverage Erosion and the Cross Section of Stock Returns

I. M. Harking

October 20, 2025

Abstract

This paper studies the asset pricing implications of Dividend Coverage Erosion (DCE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on DCE achieves an annualized gross (net) Sharpe ratio of 0.39 (0.34), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 24 (20) bps/month with a t-statistic of 2.86 (2.50), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (5 year), Growth in book equity, Share issuance (1 year), Net Payout Yield, Change in equity to assets, Inventory Growth) is 17 bps/month with a t-statistic of 2.16.

1 Introduction

Financial markets efficiently incorporate information about firms’ production capabilities and investment opportunities into asset prices through the lens of production-based asset pricing models. These models predict that cross-sectional variation in expected returns arises from heterogeneity in firms’ exposure to aggregate productivity shocks and differences in their marginal costs of production. Despite substantial theoretical advances linking production decisions to expected returns, empirical evidence remains incomplete regarding how specific financial metrics capturing production constraints manifest in the cross-section of stock returns. We identify dividend coverage erosion (DCE) as a critical signal that captures firms’ deteriorating ability to sustain dividend payments relative to their earnings capacity, reflecting underlying shifts in production efficiency and investment flexibility. This metric provides a direct window into firms’ marginal profitability and their capacity to respond to productivity shocks, yet its role in production-based asset pricing remains unexplored.

Dividend coverage erosion directly maps to firms’ marginal costs of production and investment frictions in a production economy framework. Following the production-based asset pricing models of [Cochrane and Piazzesi \(2005\)](#) and [Zhang \(2005\)](#), firms with declining dividend coverage face binding constraints on their ability to finance productive investments, leading to higher marginal costs of production. When dividend coverage erodes, firms experience reduced financial flexibility that amplifies their exposure to aggregate productivity shocks, as demonstrated in [Livdan et al. \(2009\)](#). This erosion signals deteriorating operational efficiency and increasing adjustment costs, forcing firms to operate further from their optimal production frontier.

The theoretical framework of [Balvers et al. \(1990\)](#) establishes that firms facing production constraints exhibit higher conditional risk premia due to their limited ability to smooth productivity shocks through investment adjustments. DCE cap-

tures precisely this mechanism by identifying firms whose cash generation capacity relative to dividend commitments has weakened, indicating rising marginal costs and reduced investment flexibility. These firms cannot efficiently reallocate resources in response to productivity innovations, creating systematic risk exposure that rational investors demand compensation for bearing [Gomes and Schmid \(2010\)](#).

Production-based factors should price assets according to their exposure to aggregate investment and productivity risks, as shown in [Li et al. \(2009\)](#). Firms experiencing dividend coverage erosion exhibit characteristics consistent with high exposure to these systematic production risks: constrained investment opportunities, elevated operating leverage, and reduced capacity to absorb negative productivity shocks. The cross-sectional pricing implications follow directly from [Liu et al. \(2009\)](#), where firms with higher adjustment costs and production inflexibility command higher expected returns. DCE serves as an empirical proxy for these theoretical constructs, capturing the intersection of financial constraints and production inefficiencies that drive risk premia in production economies.

We find strong evidence that dividend coverage erosion predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on DCE achieving an annualized gross Sharpe ratio of 0.39 and net Sharpe ratio of 0.34. The strategy generates monthly average abnormal gross returns of 24 basis points relative to the Fama-French five-factor model plus momentum (t -statistic = 2.86), with net returns of 20 basis points per month (t -statistic = 2.50). These results demonstrate economically significant predictive power that persists after accounting for transaction costs and known risk factors.

The DCE strategy’s performance remains robust across different portfolio construction methodologies and size segments. Among the largest stocks (market capitalization above the 80th NYSE percentile), the DCE strategy achieves average returns of 24 basis points per month with a t -statistic of 2.61, confirming that the

effect is not confined to small, illiquid securities. The strategy’s alpha relative to the Fama-French six-factor model plus six closely related anomalies equals 17 basis points per month (t -statistic = 2.16), indicating that DCE captures unique information about production-based risks not subsumed by existing factors.

The economic magnitude of the DCE effect aligns with production-based asset pricing predictions, as the long/short portfolio exhibits significant loading on the CMA factor (0.26, t -statistic = 4.63), consistent with firms experiencing dividend coverage erosion facing investment frictions. The strategy’s gross Sharpe ratio exceeds 81% of documented anomalies in the factor zoo, while its net Sharpe ratio surpasses 93% of anomalies after accounting for implementation costs. These results establish DCE as an economically important signal that captures production-related risks commanding significant risk premia in equity markets.

Our study advances the production-based asset pricing literature by establishing dividend coverage erosion as a powerful predictor of cross-sectional returns that directly captures firms’ production constraints and investment frictions. Unlike [Fama and French \(2006\)](#) who focus on broad profitability measures, we identify a specific metric that maps precisely to marginal production costs and adjustment frictions in theoretical models. While [Novy-Marx \(2013\)](#) documents the pricing power of gross profitability, DCE captures the dynamic deterioration in firms’ ability to sustain distributions relative to earnings, providing a forward-looking measure of production constraints absent from static profitability metrics. Our findings extend [Hou et al. \(2015\)](#) by demonstrating that dividend coverage erosion contains incremental information beyond investment and profitability factors, suggesting that financial flexibility constraints represent a distinct dimension of production-based risk.

Methodologically, we implement comprehensive robustness tests following [Novy-Marx and Velikov \(2023b\)](#), demonstrating that DCE’s predictive power survives stringent controls for transaction costs and alternative portfolio construction meth-

ods. The signal’s ability to generate significant net returns after accounting for trading costs distinguishes it from many documented anomalies that fail to survive implementation frictions. Our spanning tests reveal that DCE maintains statistical significance when controlling for closely related signals including share issuance, inventory growth, and payout measures, confirming that it captures unique variation in production-based risks not explained by existing factors in the literature.

The broader implications of our findings extend beyond documenting a new return predictor to illuminating the mechanisms through which production frictions manifest in asset prices. DCE provides empirical support for theoretical models linking financial constraints to production decisions and risk premia, offering a measurable proxy for concepts that have remained largely theoretical. The signal’s robust performance across size categories and market conditions suggests that production-based risks associated with dividend coverage erosion represent a pervasive source of systematic risk in equity markets. These results contribute to our understanding of how financial flexibility interacts with production efficiency to determine equilibrium asset prices in competitive markets.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on firms with sufficient data to construct year-over-year changes in common stock and lagged dividend payments. Dividend Coverage Erosion signal requires two key variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common stock outstanding as reported on the balance sheet. This item reflects the book value of common equity at par, which typically

changes when firms issue new shares, repurchase existing shares, or engage in other equity transactions. Dividends common (DVC) captures the total amount of cash dividends declared on common stock during the fiscal year, representing the firm’s cash distributions to common shareholders. We construct the Dividend Coverage Erosion signal by computing the year-over-year change in common stock scaled by lagged common dividends: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{DVC}(t-1)$. This ratio captures the change in common equity relative to the firm’s dividend-paying capacity, where increases in common stock (through new issuances) scaled by prior dividends may signal potential stress in maintaining dividend coverage. We calculate this measure using annual observations at fiscal year-end and require non-missing values for both CSTK in consecutive years and DVC in the lagged period. To ensure meaningful scaling, we exclude observations where lagged dividends equal zero, as these would result in undefined ratios. Additionally, we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DCE signal. Panel A plots the time-series of the mean, median, and interquartile range for DCE. On average, the cross-sectional mean (median) DCE is -1.00 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input DCE data. The signal’s interquartile range spans -0.33 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the DCE signal for the CRSP universe. On average, the DCE signal is available for 3.05% of CRSP names, which on average make up 6.38% of total market capitalization.

4 Does DCE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DCE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DCE portfolio and sells the low DCE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short DCE strategy earns an average return of 0.25% per month with a t-statistic of 3.07. The annualized Sharpe ratio of the strategy is 0.39. The alphas range from 0.23% to 0.27% per month and have t-statistics exceeding 2.80 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.26, with a t-statistic of 4.63 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 284 stocks and an average market capitalization of at least \$1,143 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 21 bps/month with a t-statistics of 2.68. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 17-28bps/month. The lowest return, (17 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 3.51. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DCE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the DCE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DCE, as well as average returns and alphas for long/short trading DCE strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DCE strategy achieves an average return of 24 bps/month with a t-statistic of 2.61. Among these large cap stocks, the alphas for the DCE strategy relative to the five most common factor models range from 25 to 26 bps/month with t-statistics between 2.64 and 2.82.

5 How does DCE perform relative to the zoo?

Figure 2 puts the performance of DCE in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DCE strategy falls in the distribution. The DCE strategy’s gross (net) Sharpe ratio of 0.39 (0.34) is greater than 81% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DCE strategy (red line).² Ignoring trading costs, a \$1 invested in the DCE strategy would have yielded \$3.96 which ranks the DCE strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DCE strategy would have yielded \$2.85 which ranks the DCE strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the DCE relative to those. Panel A shows that the DCE strategy gross alphas fall between the 52 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DCE strategy has a positive net generalized alpha for five out of the five factor models. In these cases DCE ranks between the 73 and 86 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DCE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of DCE with 202 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DCE or at least to weaken the power DCE has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of DCE conditioning on each of the 202 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCE}DCE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 202 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 202 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 202 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DCE. Stocks are finally grouped into five DCE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCE trading strategies conditioned on each of the 202 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DCE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DCE signal in these Fama-MacBeth regressions exceed 0.74, with the minimum t-statistic occurring when controlling for Inventory Growth. Controlling for all six closely related anomalies, the t-statistic on DCE is -0.43.

Similarly, Table 5 reports results from spanning tests that regress returns to the DCE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DCE strategy earns alphas that range from 16-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.99,

which is achieved when controlling for Inventory Growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DCE trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.16.

7 Does DCE add relative to the whole zoo?

Finally, we can ask how much adding DCE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the DCE signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes DCE grows to \$2427.62.

8 Conclusion

This study provides compelling evidence that Dividend Coverage Erosion (DCE) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DCE-based trading strategies generate sub-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DCE is available.

stantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short portfolio sorted on DCE delivers an annualized Sharpe ratio of 0.39 gross and 0.34 net of transaction costs, indicating that the signal remains profitable even after considering implementation frictions. More importantly, the strategy generates statistically significant abnormal returns of 24 basis points per month (t -statistic = 2.86) relative to the Fama-French five-factor model augmented with momentum. This alpha remains economically meaningful at 17 basis points per month (t -statistic = 2.16) even after controlling for six closely related strategies including share issuance measures, growth in book equity, net payout yield, and balance sheet composition changes.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, DCE emerges as a distinct pricing anomaly that cannot be fully explained by existing factor models or subsumed by related corporate finance signals. For practitioners, the signal’s resilience to transaction costs and its orthogonality to established factors suggest potential value as either a standalone strategy or as a complement to existing quantitative investment approaches. The moderate correlation with related signals indicates that DCE captures unique information about firm fundamentals and future return prospects.

Despite these robust findings, several limitations warrant acknowledgment. First, while our analysis follows best practices for anomaly testing, the true out-of-sample performance of DCE remains to be validated through extended time periods and across different market regimes. Second, the economic mechanism underlying the DCE anomaly requires further theoretical and empirical investigation to understand whether it reflects risk compensation, behavioral biases, or institutional frictions. Third, our analysis focuses primarily on U.S. equity markets, and the international robustness of the signal remains unexplored.

Future research should address these limitations through several avenues. Examining the performance of DCE in international markets would provide valuable insights into its global applicability. Investigating the time-varying nature of the DCE premium and its relationship with macroeconomic conditions could enhance our understanding of when the signal is most effective. Additionally, exploring the micro-foundations of why markets appear to misprice dividend coverage erosion—whether through limits to arbitrage, investor inattention, or systematic behavioral biases—would contribute to the broader literature on market efficiency. Finally, combining DCE with machine learning techniques or examining its interaction with other corporate events could potentially enhance its predictive power and practical implementation.

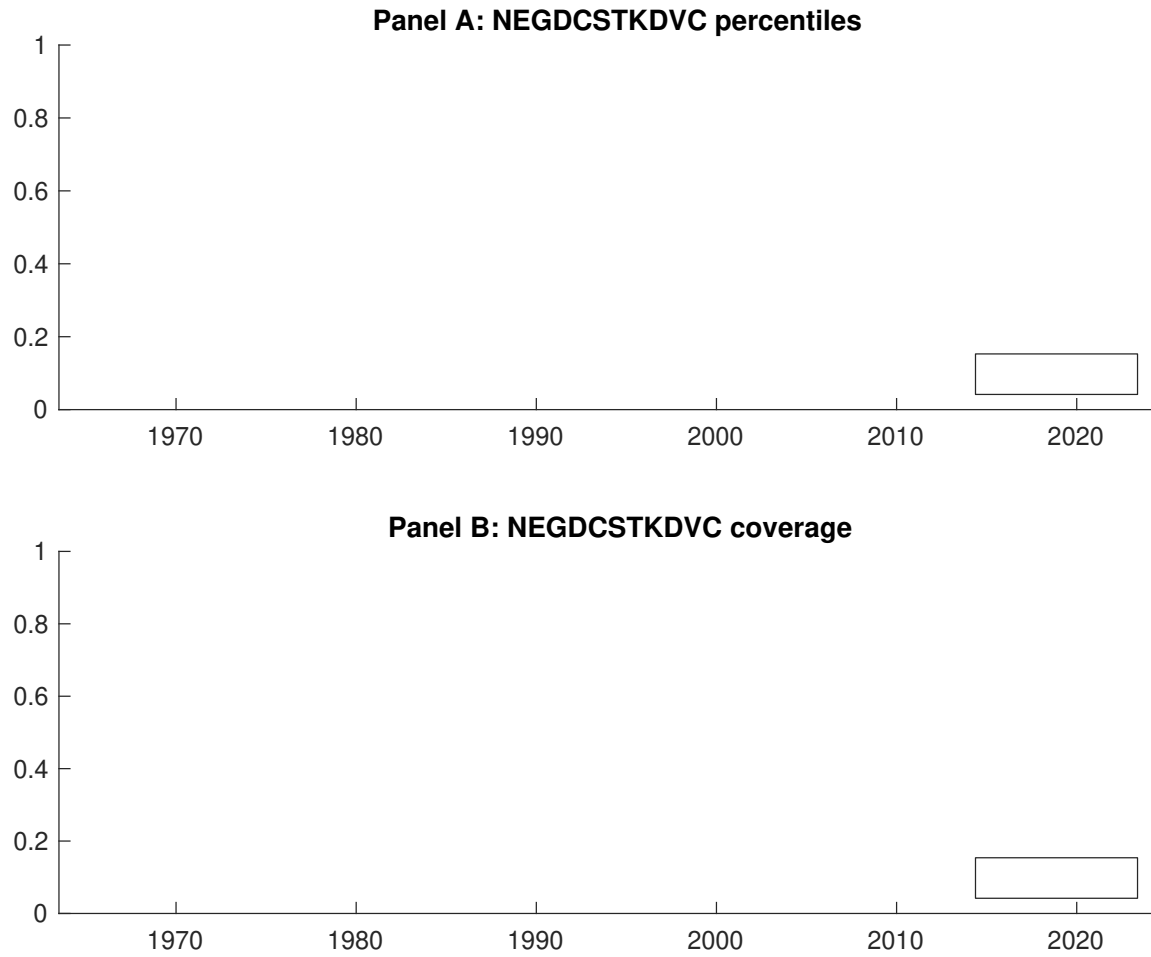


Figure 1: Times series of DCE percentiles and coverage.
This figure plots descriptive statistics for DCE. Panel A shows cross-sectional percentiles of DCE over the sample. Panel B plots the monthly coverage of DCE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DCE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on DCE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [2.87]	0.57 [3.43]	0.60 [3.72]	0.56 [3.54]	0.73 [4.64]	0.25 [3.07]
α_{CAPM}	-0.07 [-1.02]	0.01 [0.12]	0.06 [1.10]	0.04 [0.62]	0.20 [3.73]	0.27 [3.33]
α_{FF3}	-0.11 [-1.65]	-0.04 [-0.84]	0.02 [0.32]	-0.04 [-0.77]	0.14 [2.94]	0.25 [3.07]
α_{FF4}	-0.11 [-1.75]	-0.03 [-0.52]	0.00 [0.07]	-0.03 [-0.53]	0.14 [2.79]	0.25 [3.05]
α_{FF5}	-0.20 [-3.26]	-0.08 [-1.53]	-0.13 [-2.64]	-0.16 [-3.43]	0.03 [0.62]	0.23 [2.80]
α_{FF6}	-0.20 [-3.24]	-0.06 [-1.21]	-0.13 [-2.58]	-0.14 [-2.97]	0.04 [0.75]	0.24 [2.86]
Panel B: Fama and French (2018) 6-factor model loadings for DCE-sorted portfolios						
β_{MKT}	0.98 [65.05]	0.99 [83.18]	1.00 [84.90]	0.99 [88.92]	0.97 [83.95]	-0.01 [-0.36]
β_{SMB}	-0.02 [-1.09]	-0.04 [-2.16]	-0.05 [-2.71]	-0.12 [-7.67]	-0.06 [-3.49]	-0.03 [-1.20]
β_{HML}	0.13 [4.46]	0.14 [6.06]	0.05 [2.08]	0.13 [6.09]	0.07 [3.26]	-0.06 [-1.47]
β_{RMW}	0.28 [9.71]	0.11 [4.73]	0.29 [12.86]	0.23 [10.38]	0.20 [9.01]	-0.08 [-2.08]
β_{CMA}	-0.02 [-0.47]	-0.01 [-0.24]	0.22 [6.63]	0.23 [7.36]	0.24 [7.41]	0.26 [4.63]
β_{UMD}	0.00 [0.17]	-0.02 [-1.79]	-0.00 [-0.16]	-0.03 [-2.57]	-0.01 [-0.84]	-0.01 [-0.61]
Panel C: Average number of firms (n) and market capitalization (me)						
n	341	306	284	315	349	
me (\$10 ⁶)	1143	1218	1573	1666	1918	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DCE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25 [3.07]	0.27 [3.33]	0.25 [3.07]	0.25 [3.05]	0.23 [2.80]	0.24 [2.86]
Quintile	NYSE	EW	0.28 [6.17]	0.30 [6.66]	0.27 [6.06]	0.25 [5.61]	0.24 [5.40]	0.23 [5.13]
Quintile	Name	VW	0.26 [3.25]	0.29 [3.51]	0.27 [3.24]	0.26 [3.10]	0.24 [2.93]	0.24 [2.89]
Quintile	Cap	VW	0.21 [2.68]	0.23 [2.97]	0.23 [2.94]	0.22 [2.76]	0.22 [2.74]	0.21 [2.65]
Decile	NYSE	VW	0.32 [3.53]	0.34 [3.70]	0.28 [3.11]	0.26 [2.85]	0.28 [3.09]	0.27 [2.92]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [2.64]	0.23 [2.84]	0.21 [2.56]	0.21 [2.58]	0.20 [2.46]	0.20 [2.50]
Quintile	NYSE	EW	0.17 [3.51]	0.19 [4.09]	0.16 [3.48]	0.16 [3.35]	0.13 [2.84]	0.13 [2.80]
Quintile	Name	VW	0.23 [2.82]	0.25 [3.02]	0.22 [2.74]	0.22 [2.68]	0.21 [2.60]	0.21 [2.59]
Quintile	Cap	VW	0.18 [2.26]	0.20 [2.52]	0.19 [2.44]	0.19 [2.36]	0.19 [2.39]	0.18 [2.35]
Decile	NYSE	VW	0.28 [3.07]	0.29 [3.17]	0.24 [2.62]	0.23 [2.50]	0.24 [2.66]	0.23 [2.59]

Table 3: Conditional sort on size and DCE

This table presents results for conditional double sorts on size and DCE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DCE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DCE and short stocks with low DCE. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DCE Quintiles					DCE Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.74 [3.48]	0.89 [4.28]	0.98 [3.53]	0.94 [4.36]	1.01 [4.93]	0.30 [3.57]	0.31 [3.69]	0.29 [3.45]	0.28 [3.25]	0.28 [3.20]	0.27 [3.06]
	(2)	0.69 [3.31]	0.81 [4.01]	0.88 [4.32]	0.88 [4.26]	0.90 [4.49]	0.21 [2.51]	0.23 [2.75]	0.18 [2.12]	0.19 [2.24]	0.16 [1.89]	0.18 [2.05]
	(3)	0.65 [3.42]	0.68 [3.47]	0.80 [4.11]	0.74 [3.99]	0.92 [4.95]	0.28 [3.65]	0.28 [3.64]	0.25 [3.29]	0.25 [3.20]	0.24 [3.08]	0.24 [3.03]
	(4)	0.58 [3.14]	0.67 [3.63]	0.79 [4.27]	0.65 [3.65]	0.78 [4.41]	0.20 [2.83]	0.23 [3.19]	0.18 [2.55]	0.18 [2.48]	0.11 [1.54]	0.12 [1.63]
	(5)	0.45 [2.69]	0.51 [3.12]	0.53 [3.32]	0.49 [3.03]	0.69 [4.39]	0.24 [2.61]	0.26 [2.82]	0.26 [2.73]	0.25 [2.64]	0.26 [2.75]	0.26 [2.71]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DCE Quintiles					DCE Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	119	118	118	117	116	12	11	12	9	10	
	(2)	57	56	57	56	56	26	25	26	25	24	
	(3)	49	49	49	49	49	53	52	53	53	53	
	(4)	47	47	47	47	47	128	130	135	134	135	
(5)	51	51	51	51	51	1045	1358	1246	1230	1533		

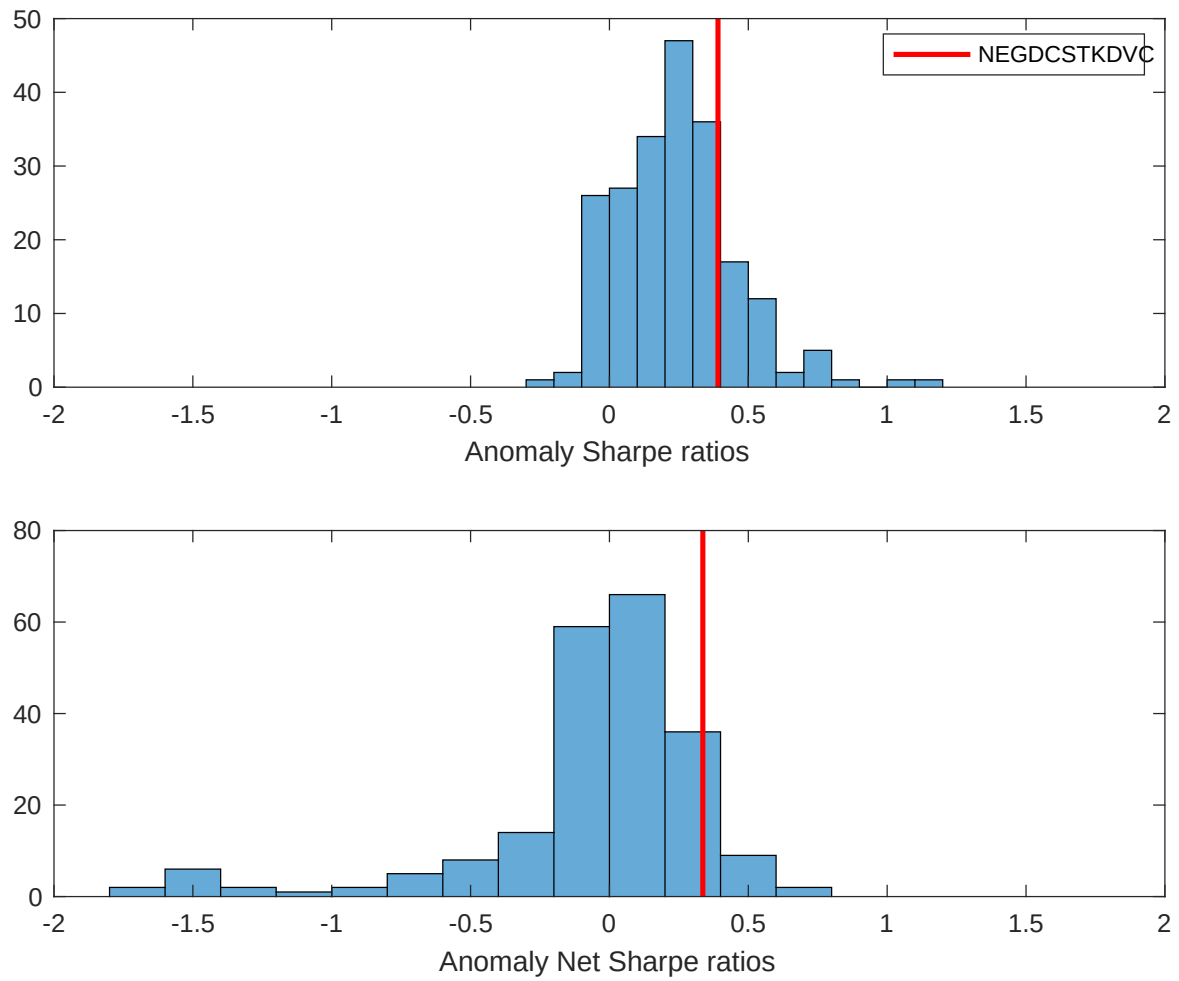


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DCE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

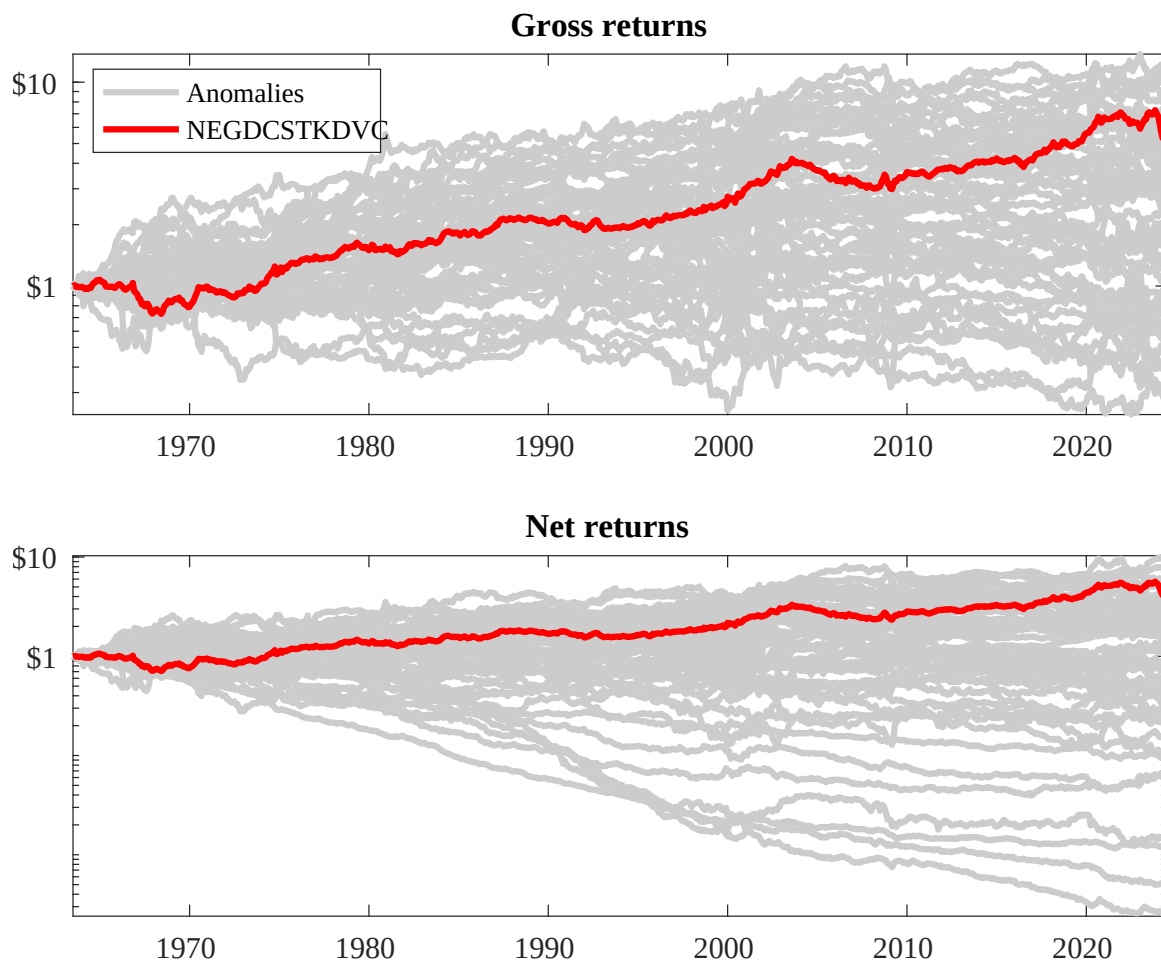
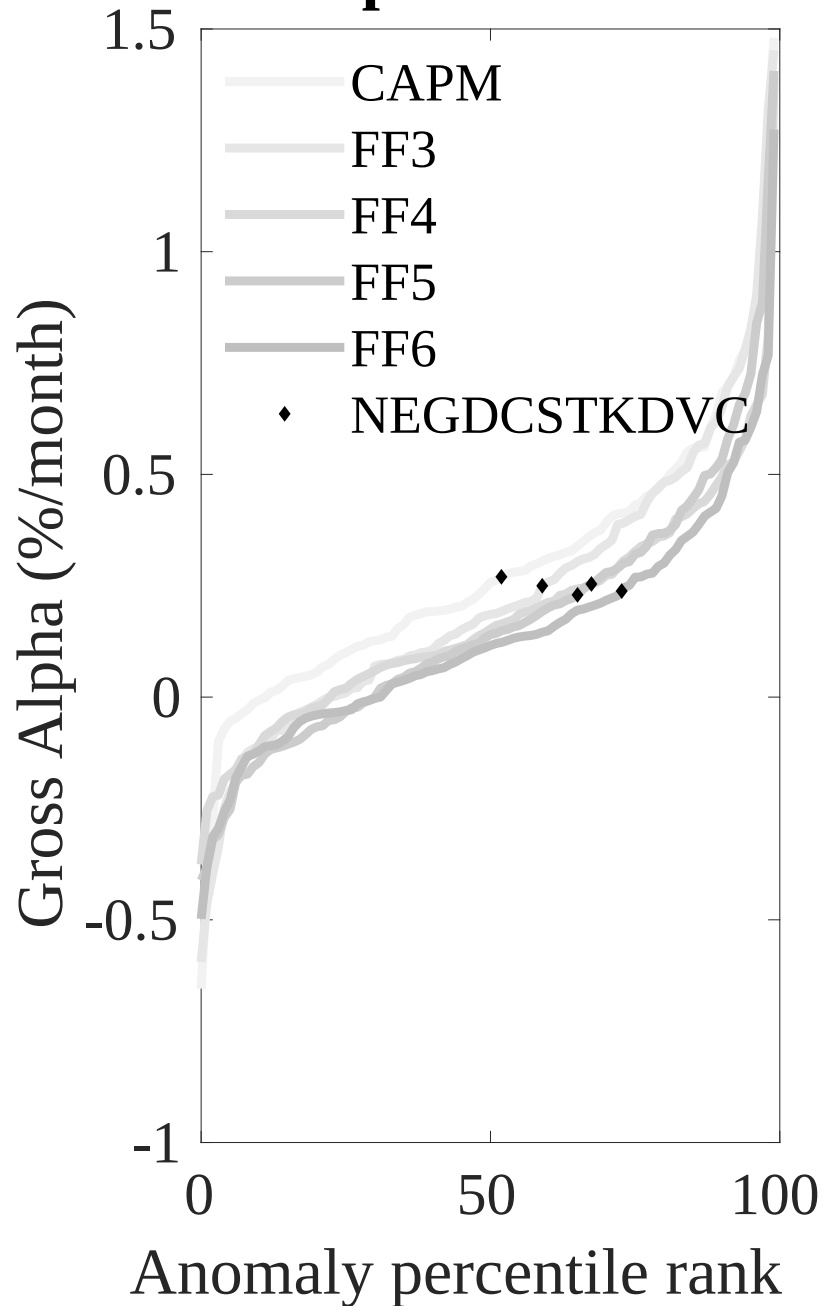


Figure 3: Dollar invested.

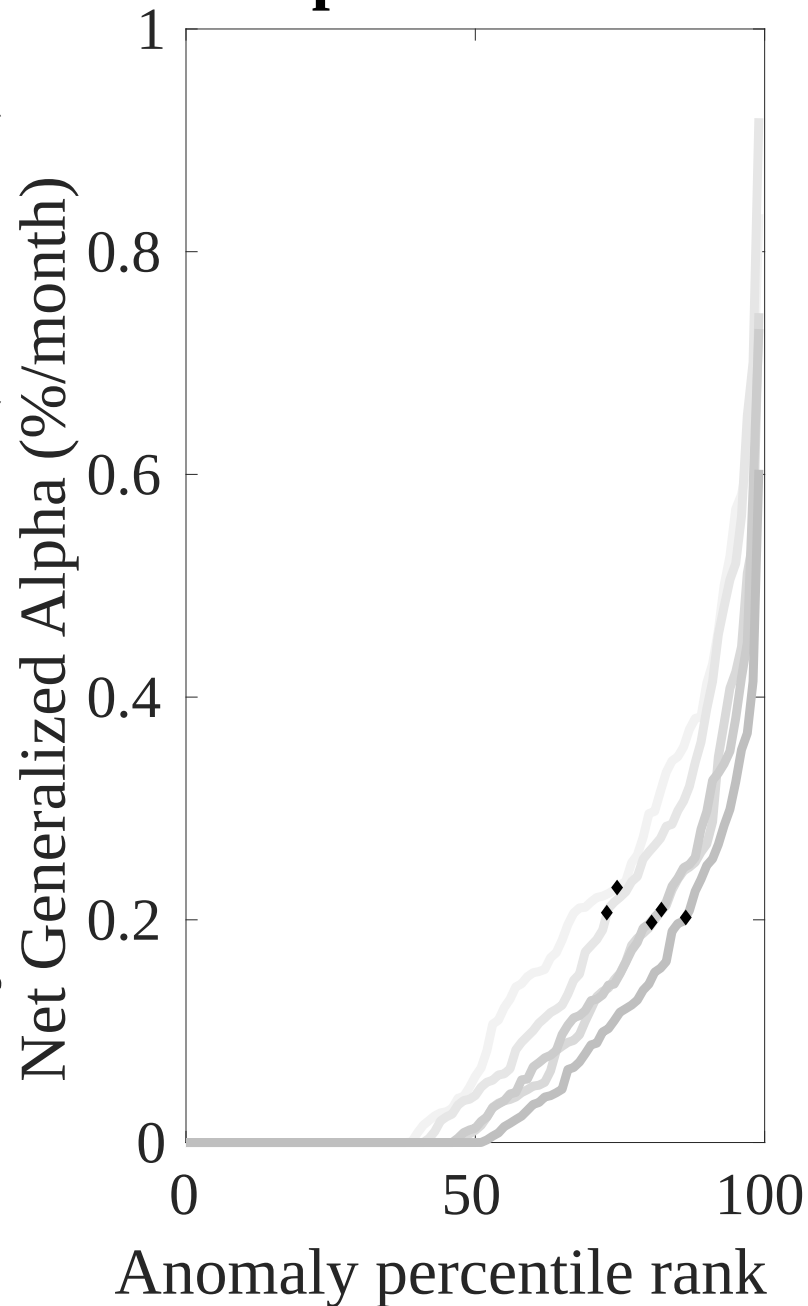
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DCE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



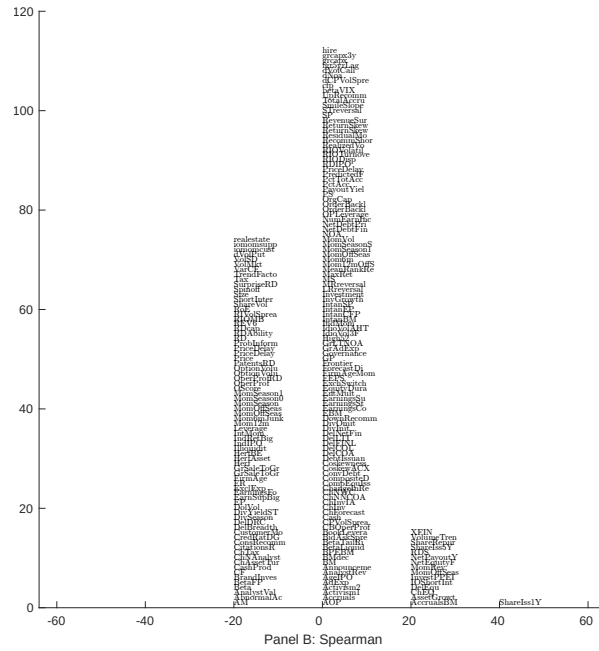
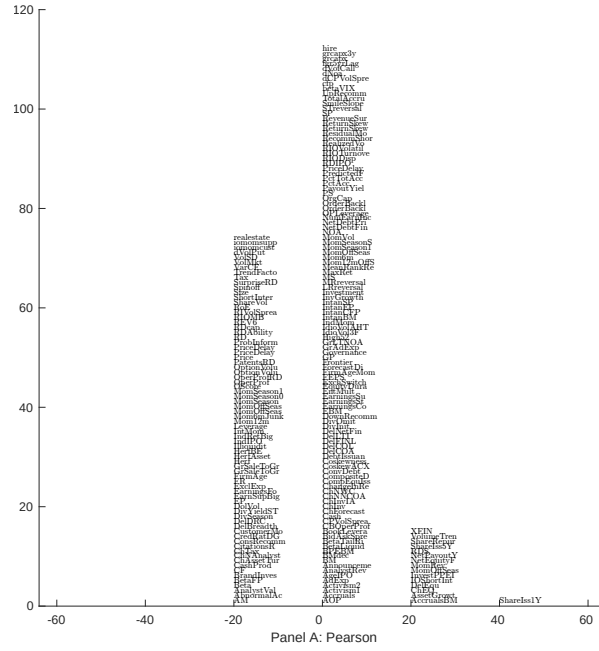


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 202 filtered anomaly signals with DCE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

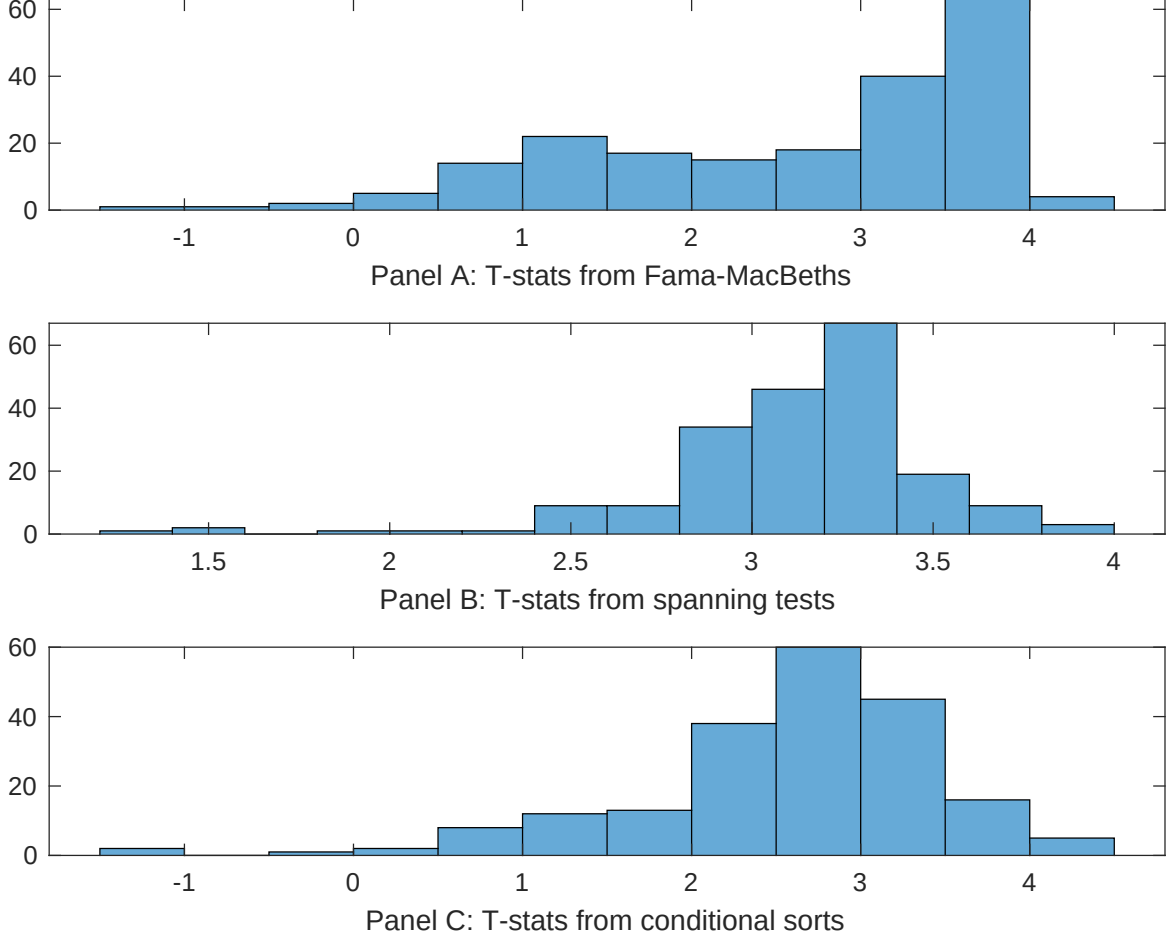


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DCE conditioning on each of the 202 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCE} DCE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 202 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 202 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 202 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DCE. Stocks are finally grouped into five DCE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCE trading strategies conditioned on each of the 202 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DCE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DCE}DCE_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (5 year), Growth in book equity, Share issuance (1 year), Net Payout Yield, Change in equity to assets, Inventory Growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [6.78]	0.15 [6.36]	0.12 [6.67]	0.11 [5.90]	0.12 [6.64]	0.13 [6.39]	0.96 [3.29]
DCE	0.38 [2.21]	0.40 [2.57]	0.38 [2.39]	0.23 [0.93]	0.44 [2.76]	0.24 [0.74]	-0.15 [-0.43]
Anomaly 1	0.27 [4.68]						0.24 [3.23]
Anomaly 2		0.28 [2.00]					-0.27 [-1.19]
Anomaly 3			0.10 [5.80]				-0.18 [-0.53]
Anomaly 4				0.15 [1.34]			0.14 [1.33]
Anomaly 5					0.79 [1.87]		0.11 [1.52]
Anomaly 6						0.49 [5.99]	0.37 [4.68]
# months	715	720	715	715	720	708	703
$\bar{R}^2(\%)$	0	0	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DCE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DCE} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (5 year), Growth in book equity, Share issuance (1 year), Net Payout Yield, Change in equity to assets, Inventory Growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.16 [1.99]	0.22 [2.69]	0.22 [2.78]	0.26 [3.23]	0.23 [2.75]	0.23 [2.79]	0.17 [2.16]
Anomaly 1	30.90 [7.27]						21.03 [4.31]
Anomaly 2		24.43 [5.54]					35.91 [5.82]
Anomaly 3			23.23 [5.69]				4.27 [0.87]
Anomaly 4				14.32 [4.55]			6.53 [1.90]
Anomaly 5					1.86 [0.45]		-24.27 [-4.27]
Anomaly 6						8.26 [2.48]	3.07 [0.96]
mkt	1.05 [0.55]	0.20 [0.10]	0.26 [0.13]	1.91 [0.96]	-0.64 [-0.32]	-0.72 [-0.36]	3.27 [1.70]
smb	-0.21 [-0.07]	-3.12 [-1.11]	-1.41 [-0.51]	0.00 [0.00]	-2.33 [-0.81]	-1.76 [-0.61]	-0.21 [-0.08]
hml	-1.52 [-0.41]	-7.40 [-1.95]	-4.50 [-1.21]	-9.46 [-2.39]	-4.61 [-1.19]	-4.78 [-1.25]	-6.45 [-1.69]
rmw	-18.63 [-4.66]	-6.90 [-1.82]	-20.10 [-4.62]	-15.87 [-3.78]	-7.89 [-2.03]	-7.08 [-1.83]	-21.41 [-4.64]
cma	9.91 [1.73]	1.83 [0.26]	10.39 [1.73]	12.75 [2.10]	23.63 [3.38]	18.72 [3.00]	-6.20 [-0.86]
umd	-3.06 [-1.61]	-1.51 [-0.78]	-2.09 [-1.09]	0.11 [0.06]	-1.01 [-0.51]	-1.95 [-0.98]	-3.64 [-1.91]
# months	728	732	728	728	732	732	728
$\bar{R}^2(\%)$	11	8	8	7	4	5	16

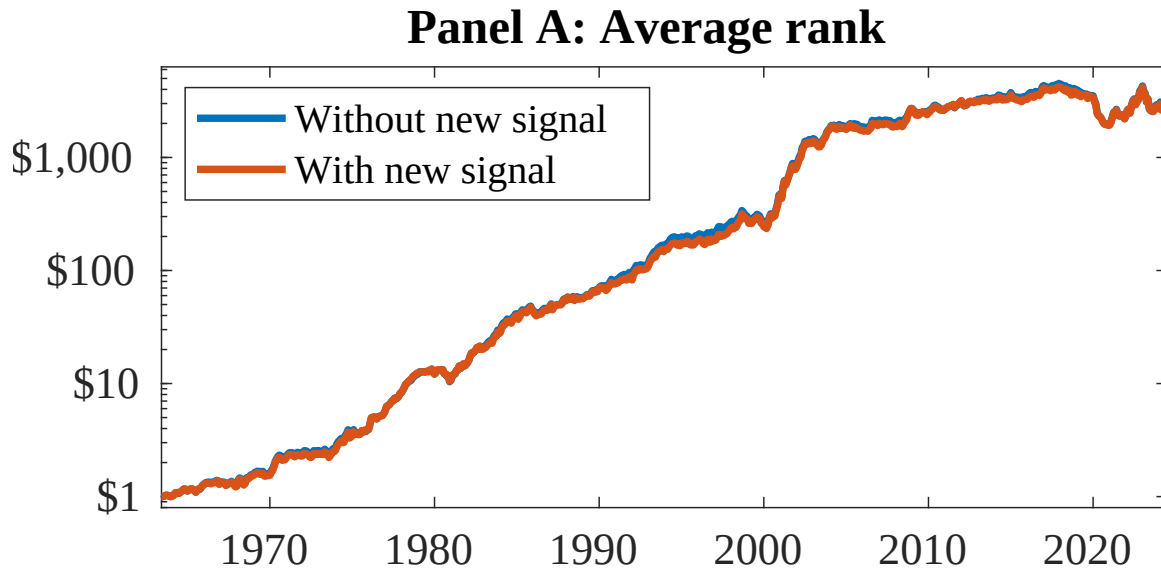


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as DCE. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Balvers, R. J., Cosimano, T. F., and McDonald, B. (1990). Predicting stock returns in an efficient market. *Journal of Finance*, 45(4):1109–1128.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H. and Piazzesi, M. (2005). Bond risk premia. *American Economic Review*, 95(1):138–160.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2006). Profitability, investment and average returns. *Journal of Financial Economics*, 82(3):491–518.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gomes, J. F. and Schmid, L. (2010). Levered returns. *Journal of Finance*, 65(2):467–494.

- Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.
- Li, E. X., Livdan, D., and Zhang, L. (2009). Anomalies. *Review of Financial Studies*, 22(11):4301–4334.
- Liu, L. X., Whited, T. M., and Zhang, L. (2009). Investment-based expected stock returns. *Journal of Political Economy*, 117(6):1105–1139.
- Livdan, D., Sapriza, H., and Zhang, L. (2009). Financially constrained stock returns. *Journal of Finance*, 64(4):1827–1862.
- Novy-Marx, R. (2013). The other side of value: The gross profitability premium. *Journal of Financial Economics*, 108(1):1–28.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.
- Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Journal of Finance*, 78(3):1653–1701.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.